

DNA and RNA sensing (NAT)

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▣ Microarrays, PCR, DNA sequencing on chip

▣ Working principles, structures on chip and fabrication

DNA Microarrays



PCR



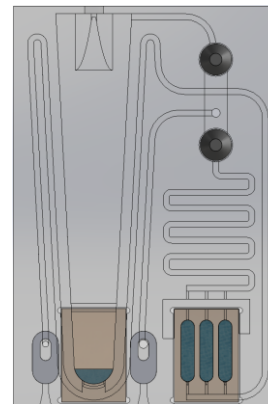
Q-PCR



DNA Sequencing



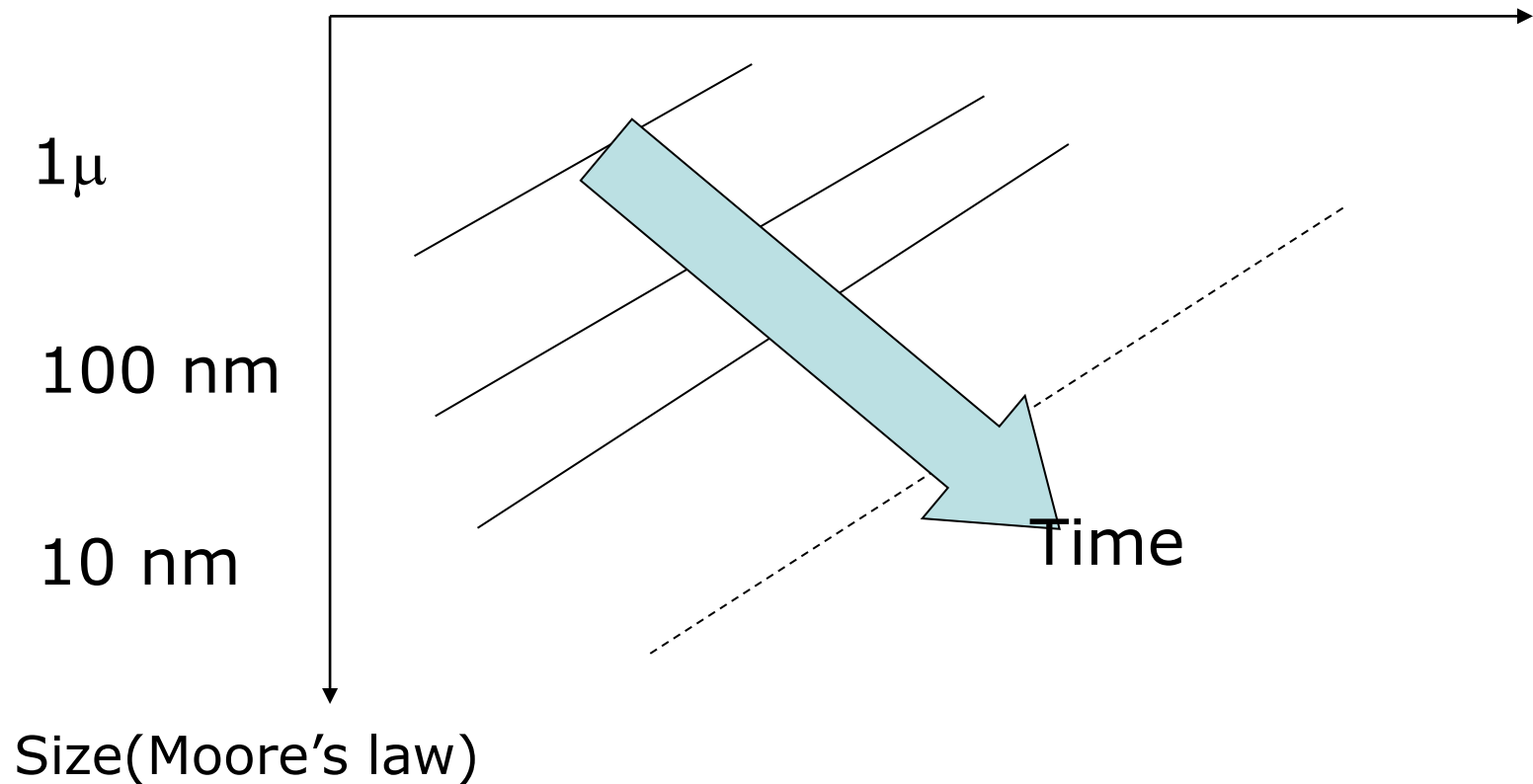
DNA purification and detection



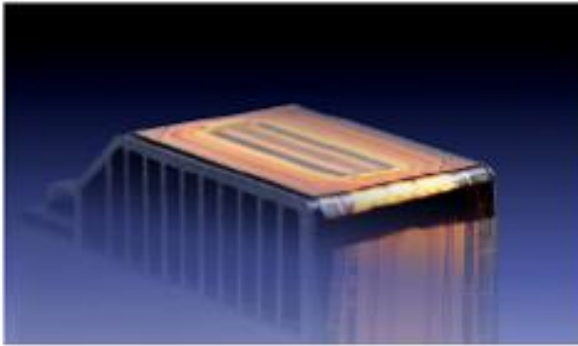
More than Moore

RF technologies on chip, High voltage and power,
Sensors and actuators on CMOS platforms, Fluidics
Optoelectronics, Image sensing

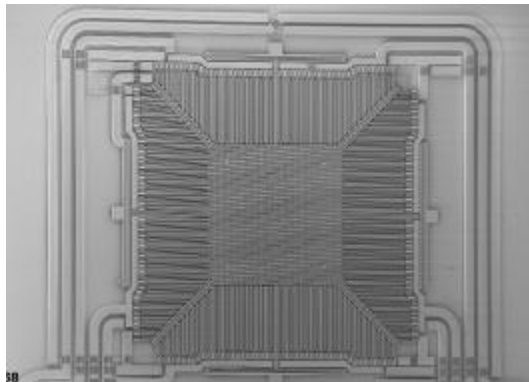
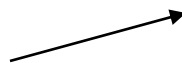
Functions



The MEMS revolution:



Micro Fluidic



Inertial sensing

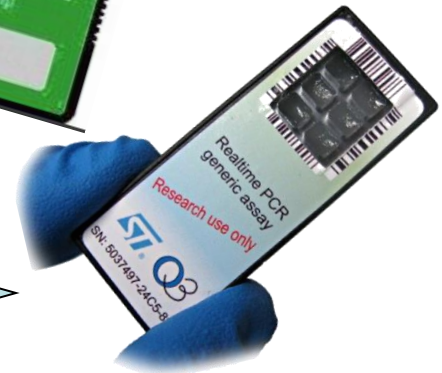
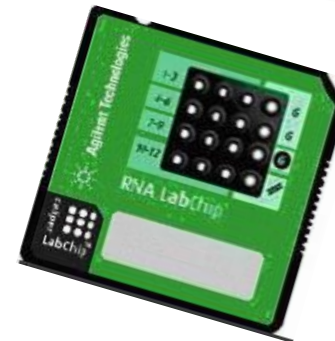
For molecular biology



From labs to "lab-on-chip"



From niche to high-volumes products



Q3 system

Instrument



Disposable cartridges

Ebola pandemic risk

Needs:

- Quickly identify ebola infected patients and separate them from others. Symptoms are easily confused with Malaria
- Control people at borders, to avoid the spreading of pandemic

Available options

- Immunoassay based tests, but are low sensitivity and not very specific
- Send sample to central labs, but in Africa are very uncommon and the infected blood is very dangerous

Advantages of NAT:

- NAT is now the gold standard for WHO, a new rapid test is straightforward

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STMicroelectronics and partners create early Ebola detector

🕒 December 16, 2014 📰 News 👤 Isaac Leung

STMICROELECTRONICS has worked with Clonit and Italy's National Institute for Infectious Diseases to create a portable analyser that detects Ebola in less than 75 minutes.

The portable analyser, currently in prototype stages, is based on the Real-Time Polymerase Chain Reaction (RT-PCR) molecular biology technique.



Polymerase Chain Reaction (PCR) amplifies a target biological material, such as the Ebola virus, contained in a tiny sample of blood so it can be detected. The key procedure in PCR is the accurately controlled cycle of repetitive heating and cooling of the biological material that is subsequently evaluated

Latest News

[HTC Launches HTC Desire 628 with JB-Hi Fi](#)

[STMicroelectronics releases low-power microcontroller series](#)

Quickly adapted for Covid19 test



6 REAZIONI
multiplex per
3 TEST in
1 CARTUCCIA



RISULTATI IN 1 ORA



**RFID:
Radio-Frequency Identification

Meat-ID

Food quality and safety

- Given a sample of minced meat, it may contain unwanted traces of animals other than the declared one.
- The final manufacturer wants to ensure quality and avoid bad reputation if fraud is discovered. (in bonafide)
- Today the internal quality-control labs of food industries does not perform genetic tests. Big companies may invest in that, small producers cannot afford it.
- Meat-ID tm measures contents of Beef, Pork, chicken and horse. It also detect a gene of high valuable "piemontese" beef
- Applications are:
 - Avoid intentional/unwanted frauds
 - Guarantee genuine "piemontese" beef
 - Halal / kosher food certification (100% pork free)



XY Test

Just for demo purpose

- Provide a simple example of a complete test flow on a trivial case: gender identification
- Principle of operation
 - Humans have 23 chromosomes pairs, 22 pairs are homologues, the last pair is different in males and females
 - In females, the last pair is composed by two chromosomes of the same type, named "X" chromosomes, so Females have a "XX" pair
 - Males also have one "X", but the other in the pair is a different type, named "Y", so males have an "XY" pair.
 - Being able to detect the presence of "X" and "Y", one can discriminate if the sample comes from a male or a female
- Application:
 - The chip already contains all the reagents for the test
 - The sample is an eluted buccal swab
 - The software runs on an Android device, connected via Bluetooth.
 - It takes ~40 min to perform



Which will be the successful application?

"Prediction is very difficult, especially if it's about the future."

--Nils Bohr, Nobel laureate in Physics

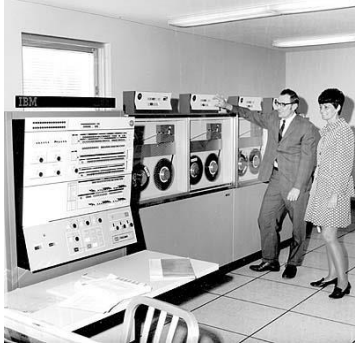
"It is far better to foresee even without certainty than not to foresee at all. "

--Henri Poincare in The Foundations of Science.

"The only way we can guess about the future in a certain situation is to look at the past, and with similar starting conditions expect a similar outcome. "

IT paradigm

Mainframe



- Research
- Banking
- Enterprise
- ...

~ 1960 1980

Small
business



Personal Computer



- Desktop document creation
- Office automation
- e-mail

~ 1980 1995

Internet



Multimedia

Portable appliances



- Entertainment:
 - Games
 - Music
 - Movies
- Communication:
 - Voip
 - Blogs
 - Social NW

Cost	>100k\$	>10k\$	<1k\$
Target	Professionals	Experts	Everybody

Biotech Paradigma

Labs



- Research
- Hospital
- Biotech enterprises

→
Small
health
centers

Today

Lab on chips



- ? Food control
- ? Environment



Personal
Genetics ?



- ???
- ???
- ???

?

Cost	> 100k\$	> 10k\$	< 1k\$
Target	Professionals	Experts	Everybody

Consumer genetics



Now, you can find out if you're having a [Boy or a Girl](#) as early as week 7 during your pregnancy. See our [Pink or Blue® Gender Test](#) for more information.



Recent discoveries show that the amount of [FAEE found in hair](#) correlates with amount of alcohol consumption over time. Read more: [AbuseCheck™ Alcohol & Drug Abuse Test](#).



[Paternity](#) is a commonly asked question. Today's technologies provide accuracies of greater than 99%. See more about [DNA Paternity Testing](#).



Personalized medicine is now a reality. Genes affect the way individuals respond to many medications. Read more about the [AsthmaGEN™ DNA test](#).



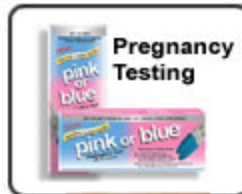
Your current caffeine intake may put you at risk of heart attack or decrease your fertility. See our [Caffeine Metabolism DNA test](#) to learn more.



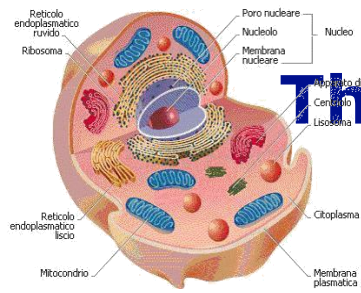
Genetic studies have revealed that, for some people, drinking moderately can lower cholesterol levels and risk of heart attacks. See more about [Alcohol Metabolism](#) testing.



Today, many people are using DNA to verify biological relationships, such as [siblingship](#), [grandparentage](#), or [twin zygosity](#).



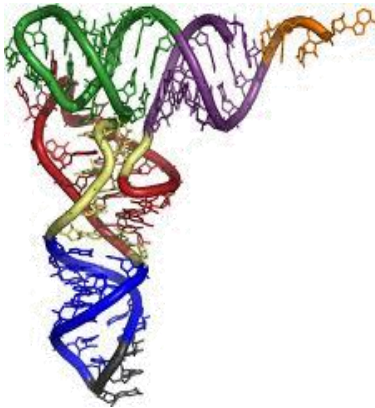
The [Pink or Blue Pregnancy Test w/ Early Gender Kit](#) is used to first test for pregnancy and then designed to be used to determine gender shortly after a positive pregnancy test.



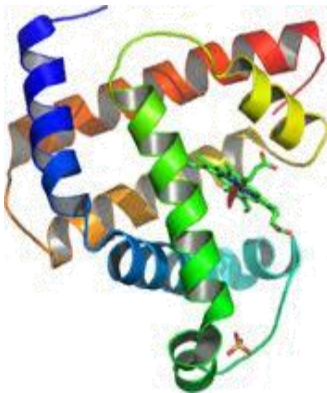
The cell works like a computer



DNA is the mass storage



RNA is the working memory



Proteins are the output



Point of care solutions



Abbot i-Stat



Abaxis "Piccolo xpress"



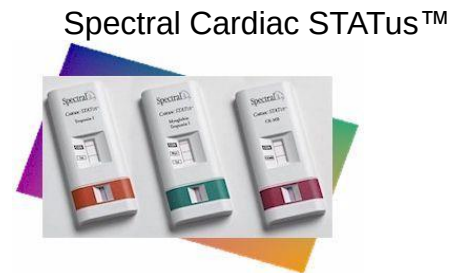
QuickVue



Biosite "triage"



IN Ratio



Spectral Cardiac STATus™



HemoCue:

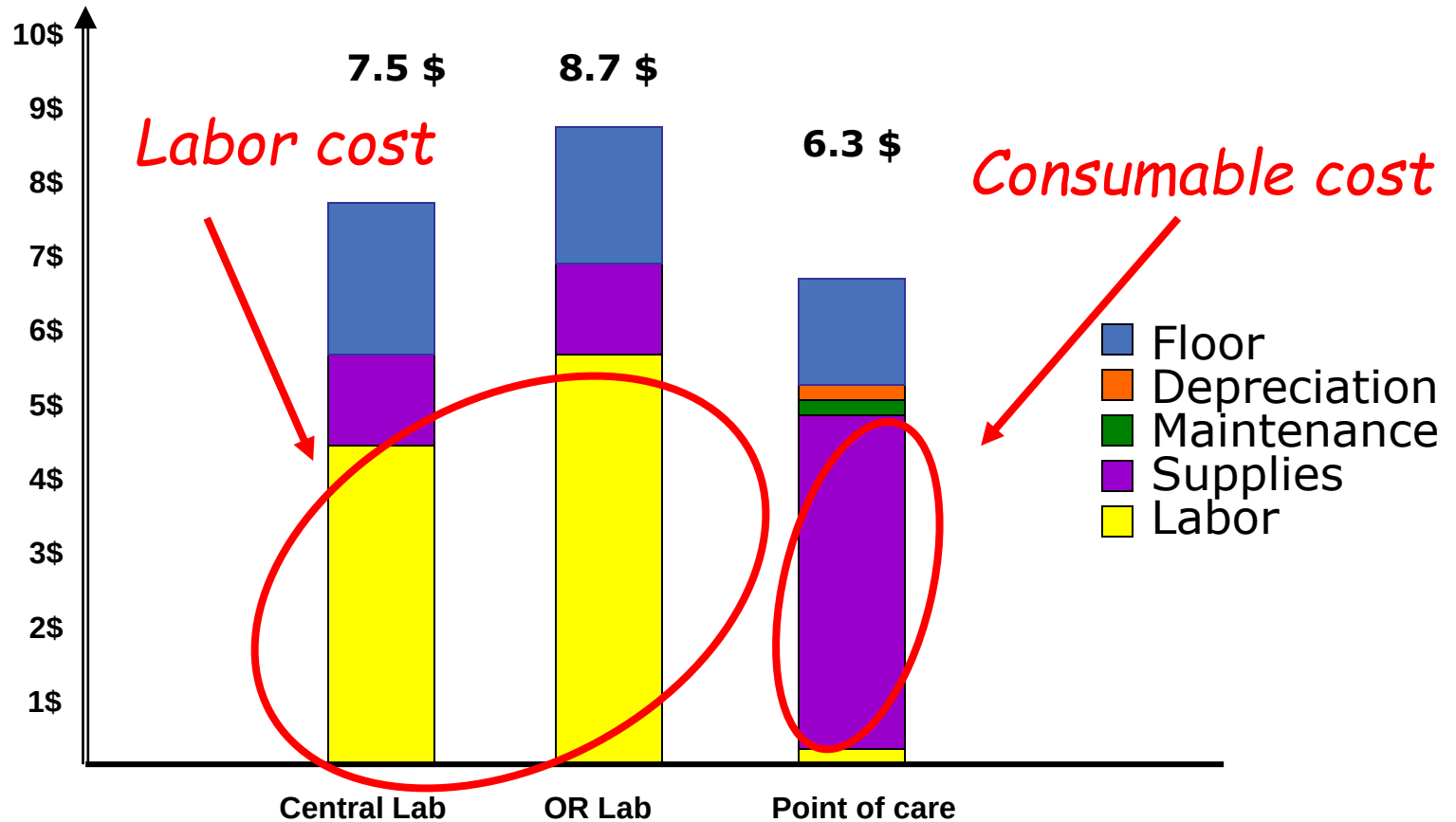
- Hemoglobin
- Plasma
- Glucose
- Albumin



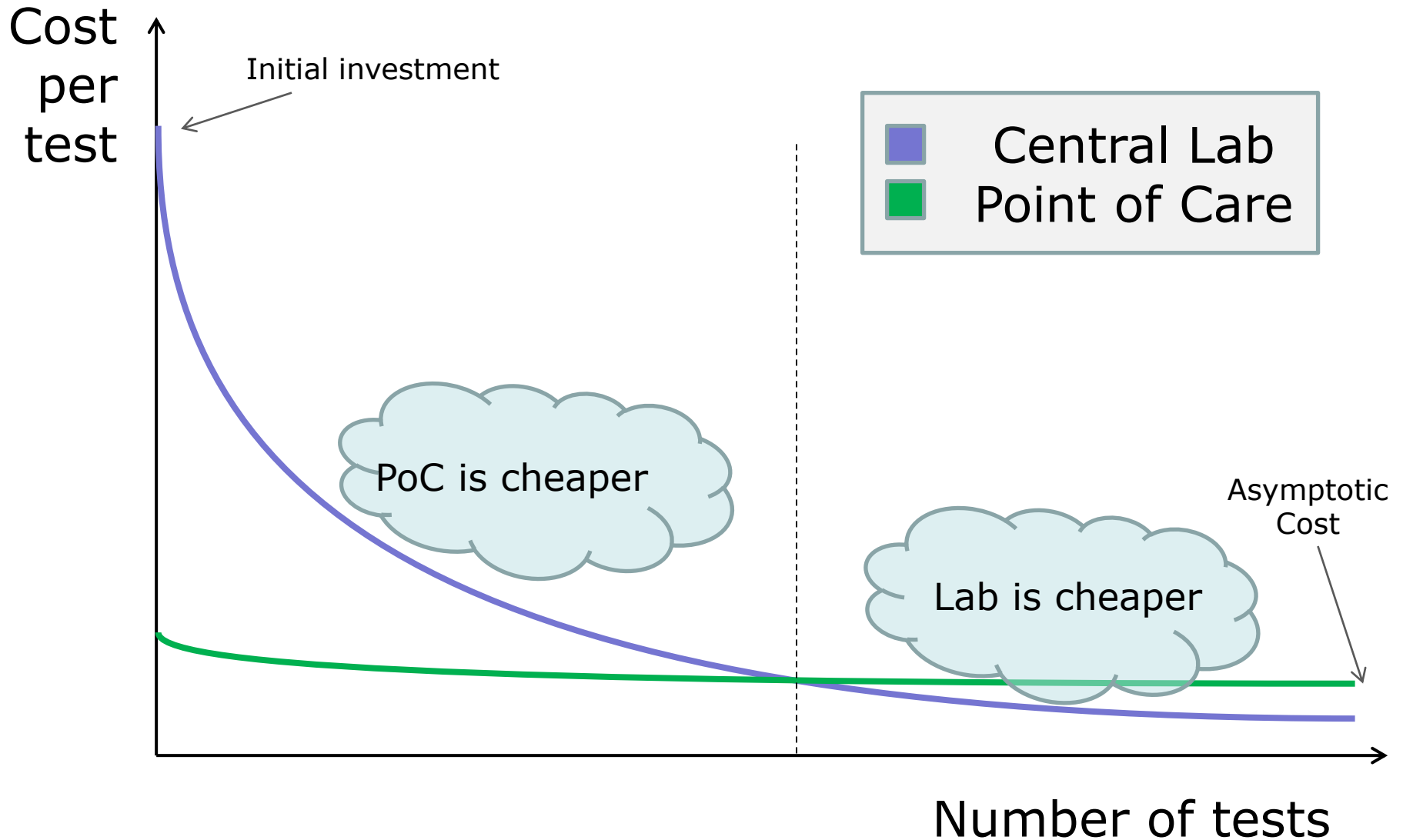
Genzyme "Osom"

Point-of-care cost per test

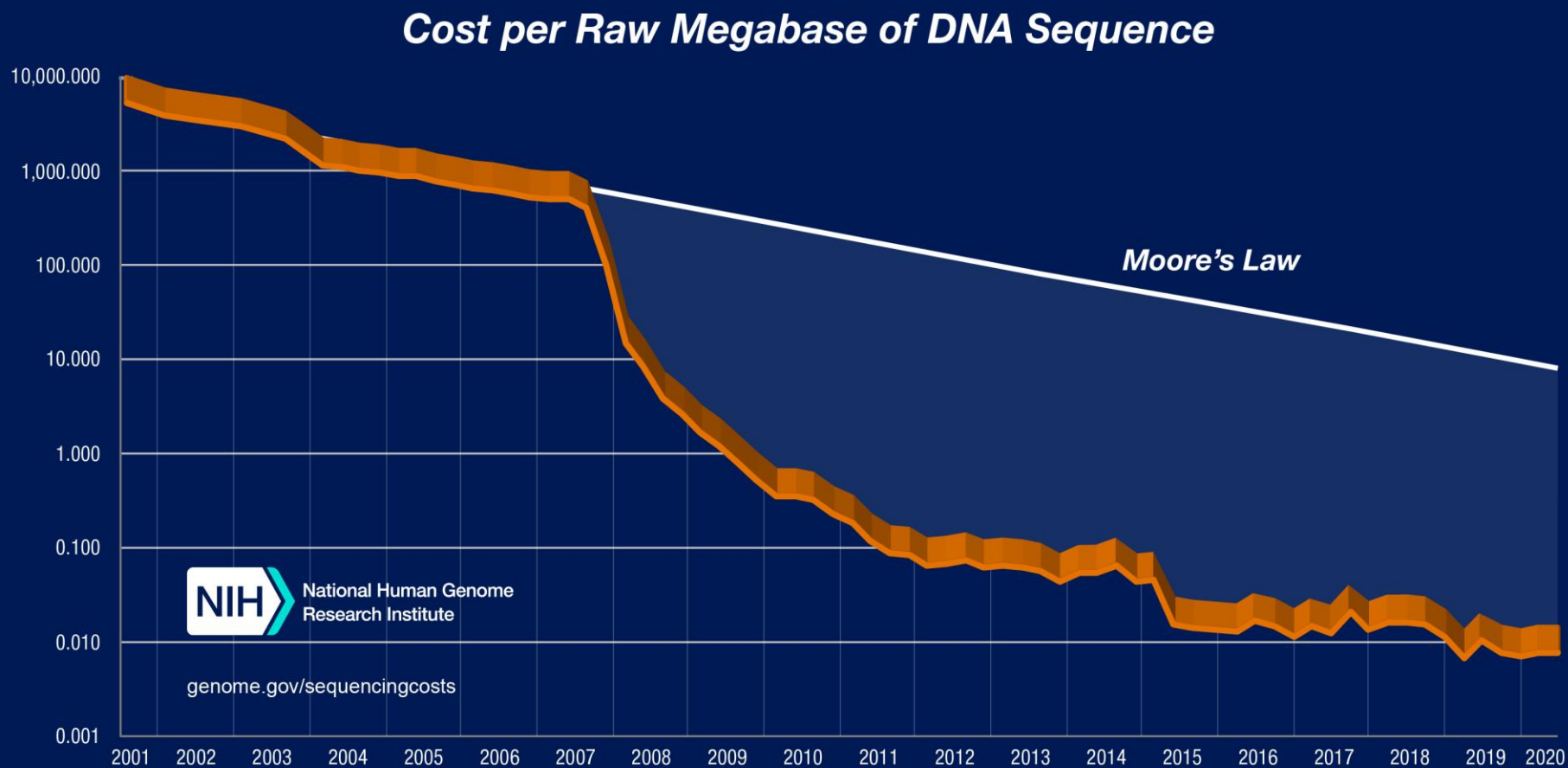
Example: gas contents in blood



Cost comparison



Sequencing cost





Scared by biotechnology?

▣ Main fears from biotech are:

▣ Individuals discrimination

▣ Creation of dangerous GMO/viruses lethal to humanity



▣ Old fears from IT were:

▣ Artificial super intelligence to control our lives

▣ Risks of placing sensitive information online



▣ Even older

▣ Electricity and nuclear power

